HOSPITAL PATIENT DATA ANALYSIS AND RISK PREDICTION USING DATA SCIENCE

Interactive Thyroid and Cardiovascular Data Analytics Dashboard

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***Abstract*—The increasing availability of healthcare-related data creates opportunities to apply data science techniques for discovering patterns, identifying relationships, and summarizing complex information. However, meaningful analysis requires appropriate data preprocessing, statistical methods, visualization techniques, and analytical tools. The ThyroidLab project addresses this requirement by developing an interactive data analytics dashboard using the R programming language and the Shiny framework.**

ThyroidLab is designed to explore thyroid biomarkers and cardiovascular-related variables through a unified web-based interface. The system examines variables such as age, gender, body mass index (BMI), thyroid-stimulating hormone (TSH), free triiodothyronine (Free T3), free thyroxine (Free T4), thyroid peroxidase antibodies (TPO), cholesterol, blood pressure, diabetes, and smoking status. These variables are used to demonstrate how demographic information, laboratory measurements, and categorical attributes can be analyzed together.

The application integrates multiple stages of the data science workflow, including data generation and import, data normalization, exploratory data analysis, data quality assessment, preprocessing, descriptive statistics, correlation analysis, hypothesis testing, and machine learning. Users can investigate numerical distributions, compare groups, examine relationships between variables, and explore multivariate patterns using interactive visualizations. The system also supports CSV upload, filtering, sorting, data export, automated insights, and PDF report generation.

The dashboard incorporates logistic regression, decision tree, and random forest classifiers for educational model evaluation. It also provides separate illustrative risk estimators for thyroid-related and cardiovascular outcomes. These features demonstrate how statistical computing and machine learning can be integrated into a practical analytical application.

The project is implemented using R, Shiny, ggplot2, Plotly, DT, and relevant statistical and machine learning packages. Its default dataset consists of 600 synthetically generated records, allowing users to explore the application without requiring an external dataset. Since these records are synthetic and the illustrative models are not clinically validated, the application is intended for academic demonstration and data science learning rather than medical diagnosis or treatment.

2. INTRODUCTION

2.1 Background

Data science is an interdisciplinary field that combines programming, statistics, data processing, visualization, and machine learning to extract useful information from datasets. Modern analytical applications must do more than store data: they should help users understand distributions, identify unusual observations, compare groups, and investigate relationships between variables.

Healthcare-related data provides an important context for demonstrating these techniques because it may contain numerical measurements, demographic information, laboratory values, and categorical outcomes. Analyzing these attributes together allows

users to understand how different data science methods operate on a structured dataset.

For example, a dataset containing age, BMI, blood pressure, thyroid hormone measurements, cholesterol levels, and smoking status can be examined using histograms, scatter plots, correlation coefficients, group comparisons, and classification algorithms. Each method provides a different perspective on the data.

2.2 Problem Statement

Traditional data analysis often requires users to write separate scripts for importing datasets, cleaning records, generating visualizations, performing statistical tests, and evaluating machine learning models. This approach can make analysis time-consuming, particularly for users who need to experiment with different variables or visualization techniques.

Static charts also limit interactive exploration because users cannot easily change variables, apply filters, compare groups, or inspect individual observations. Furthermore, preprocessing operations and analytical results may be difficult to organize when different tools are used for each stage.

Therefore, there is a need for an integrated analytical platform that combines data management, preprocessing, visualization, statistical analysis, and machine learning within a single interface.

2.3 Proposed Solution

ThyroidLab provides a modular, interactive dashboard that brings these analytical capabilities together using R Shiny. Users can work with the built-in synthetic dataset or upload a compatible CSV file. The application provides separate modules for distribution analysis, relationship exploration, multivariate visualization, statistical summaries, data quality, statistical tests, machine learning, risk estimation, and report generation.

The reactive nature of Shiny allows the interface to update analytical outputs when users change variables, chart types, grouping options, or filters. This creates a more flexible environment for understanding data science methods and examining the results of different analytical choices.

The proposed system focuses on accessibility, modularity, reproducibility, and integration of analytical techniques. It demonstrates an end-to-end workflow without requiring users to run a separate script for every analysis.

2.4 Scope of the Project

The project focuses on educational data analytics involving thyroid biomarkers and cardiovascular-related variables. Its scope includes:

- Importing, normalizing, filtering, and examining structured datasets.
- Exploring numerical distributions and categorical compositions.
- Studying associations between numerical variables and comparing groups.
- Assessing data completeness and identifying missing values, duplicates, and potential outliers.
- Applying configurable data preprocessing techniques.
- Performing descriptive statistics, correlation analysis, and hypothesis testing.
- Demonstrating binary classification and model evaluation.
- Generating interactive visualizations, analytical summaries, and downloadable reports.

3. OBJECTIVES

The primary objective of ThyroidLab is to develop an integrated, interactive platform that demonstrates the practical application of data science programming techniques to structured thyroid and cardiovascular-related data.

The specific objectives are as follows.

1. Interactive dashboard development: To build a user-friendly analytical interface using R Shiny, with separate modules for different data science tasks.

2. Data exploration: To examine numerical variables, categorical attributes, frequency distributions, central tendencies, spread, and unusual observations.

3. Statistical visualization: To implement histograms, density curves, box plots, violin plots, empirical cumulative distribution functions, Q–Q plots, scatter plots, regression plots, heatmaps, and multivariate visualizations.

4. Relationship analysis: To investigate statistical associations between numerical measurements and compare patterns across selected categories.

5. Data quality assessment: To identify missing values, duplicate records, and potential outliers before analysis.

6. Data preprocessing: To support missing-value treatment, duplicate removal, outlier flagging or handling, and numerical scaling.

7. Statistical analysis: To calculate descriptive statistics and demonstrate correlation tests, independent-group comparisons, analysis of variance, non-parametric tests, and chi-square tests.

8. Machine learning: To train and evaluate logistic regression, decision tree, and random forest classifiers using configurable predictors and training/test splits.

9. Interactive data management: To allow users to upload compatible CSV files, filter records, inspect tables, and download selected or processed data.

10. Analytical reporting: To generate automated insights, statistical summaries, PDF reports, text summaries, and cleaned CSV exports.

11. Reproducibility: To support repeatable demonstrations through configurable random seeds and clearly defined analytical settings.

12. Practical learning: To demonstrate how R programming, statistical methods, interactive visualization, and machine learning can work together in a complete data analytics workflow.

4. NOVELTY

The main novelty of ThyroidLab lies in the integration of multiple data science operations into a single interactive dashboard. Instead of requiring separate programs for data cleaning, statistical visualization, hypothesis testing, and machine learning, the application provides a unified interface for performing these tasks.

4.1 Unified Analytical Environment

The application combines exploratory data analysis, preprocessing, statistical testing, multivariate visualization, and classification in one system. This allows users to move from inspecting raw records to examining statistical patterns and model-evaluation results within the same workflow.

4.2 Configurable Visual Analytics

Users can select variables, chart types, grouping categories, and display options. The Distribution Lab supports multiple distribution techniques, while the Relationship Lab enables users to change the X-axis and Y-axis variables and investigate relationships through different plotting methods.

This flexibility makes the application useful for comparative exploration rather than limiting users to predefined static charts.

4.3 Integrated Data Quality Management

The Data Quality module provides configurable controls for missing-value treatment, duplicate removal, outlier detection, and numerical scaling. It also provides a preprocessing log and a cleaned-data preview, helping users understand which transformations have been applied to the analysis dataset.

The application separates the cleaned analysis data from the source data so that preprocessing can be examined and reset.

4.4 Combined Statistical and Machine Learning Analysis

ThyroidLab combines traditional statistical analysis with machine learning evaluation. Users can examine descriptive statistics, correlations, group differences, and categorical associations before exploring classification models.

The Machine Learning module supports logistic regression, decision trees, and random forests, with outputs such as confusion matrices, ROC curves, feature-importance visualizations where supported, and cross-validation summaries where implemented.

4.5 Multivariate Visualization

The application extends beyond basic charts by including a correlation heatmap, correlation ranking, parallel coordinates, radar profiles, and a pairwise scatter matrix. These visualizations help users examine several measurements simultaneously and investigate potential patterns across groups.

4.6 Automated Insights and Report Generation

The Insights and Report module brings analytical summaries and export facilities together. Users can obtain automated observations and download reports or datasets for documentation and further examination.

4.7 Reproducible Educational Demonstration

The default dataset is generated using a fixed random seed, and machine learning experiments include a configurable random seed. These features support repeatable demonstrations under the same settings.

The combination of these capabilities provides a practical educational environment for understanding how different data science techniques can be organized into a single application.

5.METHODOLOGY & SYSTEM ARCHITECTURE

5.1 Overall Methodology

The project follows a structured data science methodology consisting of data input, normalization, exploration, preprocessing, statistical analysis, machine learning, visualization, and reporting.

The methodology is designed to demonstrate how data moves through the application, how analytical transformations are applied, and how the resulting information is presented to the user.

Stage 1: Data Generation and Import

The system includes a built-in synthetic dataset containing 600 records. Each record includes demographic information, thyroid biomarkers, cardiovascular measurements, and selected categorical attributes.

The dataset contains fields such as patient ID, age, gender, BMI, TSH, Free T3, Free T4, TPO antibodies, HDL cholesterol, total cholesterol, systolic blood pressure, diastolic blood pressure, diabetes, smoking, thyroid status, and a cardiovascular outcome label.

Users can alternatively upload a CSV file containing the columns required by the application. The imported dataset is checked for required columns before it is accepted.

Stage 2: Data Normalization

The normalization stage standardizes supported column names and converts relevant variables into appropriate data types. Numerical fields are converted into numeric representations, while selected outcome and category fields are normalized into consistent labels.

This stage improves compatibility between imported datasets and the application's analytical modules. The uploaded data must follow the required dataset structure; normalization does not automatically make arbitrary CSV files compatible.

Stage 3: Exploratory Data Analysis

Exploratory data analysis is performed to understand the structure, distribution, and relationships within the active dataset.

The Overview module provides key dataset metrics and an initial preview. The Distribution Lab allows users to examine the distribution of numerical variables using histograms, density curves, box plots, violin plots, ECDFs, Q–Q plots, rug plots, and frequency polygons.

The Relationship Lab investigates associations between pairs of numerical variables. Users can select grouping variables to compare observations according to attributes such as thyroid status, cardiovascular outcome, diabetes, smoking, or gender.

Stage 4: Data Quality Assessment and Preprocessing

The Data Quality module allows users to examine missingness, identify duplicate records, and detect potential outliers.

The application supports three missing-value options: retaining missing values, removing rows containing missing values, or imputing numerical values with the median and categorical values with the mode.

Duplicate records can be removed through a configurable control. Outliers can be identified using the interquartile range (IQR) rule or the Z-score rule. Depending on the selected method, outliers can be flagged, replaced with boundary values through winsorization, or removed.

Numerical variables can also be scaled using min–max normalization or Z-score standardization.

The application provides a preprocessing log and a cleaned-data preview. Users can reset the cleaned dataset and download the processed records.

Stage 5: Statistical Analysis

The statistical analysis stage summarizes the data and investigates relationships between variables.

Descriptive statistics include the mean, median, standard deviation, quartiles, and range. Correlation analysis examines the direction and strength of numerical associations using Pearson or Spearman correlation, depending on the selected configuration.

Group-comparison analysis supports tests such as Welch's independent t-test, Mann–Whitney U test, one-way ANOVA, and Kruskal–Wallis. A chi-square test of independence is available for examining associations between categorical variables.

These tests help users investigate whether observed patterns warrant further analysis. Statistical significance alone does not establish causation or clinical importance.

Stage 6: Machine Learning and Illustrative Risk Estimation

The Machine Learning module allows users to select a binary classification target, predictor variables, training-data percentage, random seed, and classifier.

The available classifiers are logistic regression, decision tree, and random forest. The application divides data into training and test subsets and presents evaluation outputs, including a confusion matrix and ROC curve. Additional feature-importance and cross-validation outputs are provided where supported by the selected model and implementation.

The Risk Estimator module contains two separate illustrative estimators. The cardiovascular estimator uses age, BMI, systolic blood pressure, HDL cholesterol, diabetes, and smoking. The thyroid estimator uses age, TSH, Free T3, Free T4, and TPO antibodies.

These estimates are demonstrations of analytical programming and logistic-regression-based probability calculations. They are not validated clinical risk scores.

Stage 7: Visualization, Insights, and Reporting

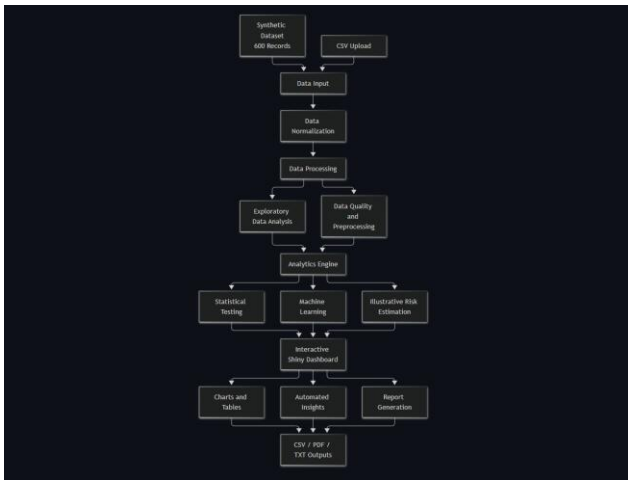
The results are displayed through interactive charts, data tables, statistical summaries, and model-evaluation panels.

The Insights and Report module provides automated analytical observations, a dataset snapshot, and download options for PDF reports, text insights, and cleaned CSV data.

This stage completes the workflow by making analytical results available for review, documentation, and further exploration.

5.2 System Architecture

ThyroidLab follows a modular, reactive architecture implemented using R Shiny. The system consists of six logical layers.



1. Data Input Layer

This layer accepts the built-in synthetic dataset or a compatible CSV uploaded by the user. It provides the initial source data for subsequent operations.

2. Data Processing Layer

This layer performs supported column-name normalization, data-type conversion, filtering, missing-value treatment, duplicate removal, outlier handling, and numerical scaling.

3. Analytics Layer

This layer contains the analytical operations, including exploratory data analysis, descriptive statistics, correlation analysis, hypothesis testing, classification, and illustrative risk estimation.

4. Visualization and User Interface Layer

The Shiny interface presents interactive charts, tables, controls, statistical summaries, and model-evaluation outputs. Plotly supports interactive charts, ggplot2 provides plotting functionality, and DT supports interactive tables.

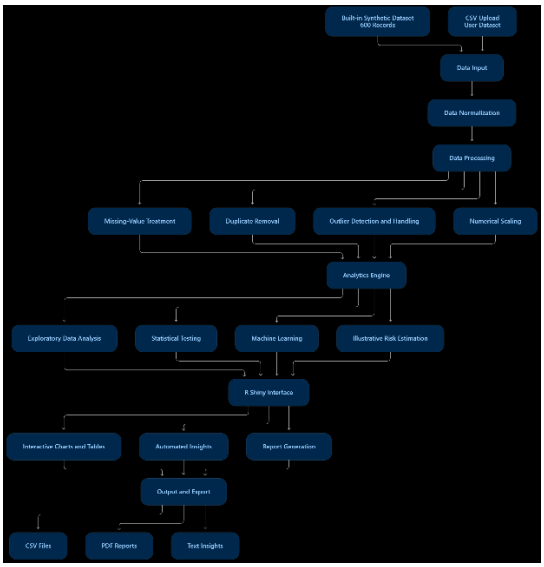
5. Insight and Reporting Layer

This layer provides automated insights, report snapshots, and facilities for generating downloadable analytical outputs.

6. Output Layer

The final outputs include interactive visualizations, statistical summaries, model-evaluation results, filtered CSV files, cleaned CSV files, text insights, and PDF reports.

System Workflow



The workflow is implemented through reactive components rather than a strictly one-way pipeline. For example, changing a chart selection updates its visualization, while changing filters affects the active dataset used by dependent outputs.

5.3 Technologies Used

R Programming Language: Used for dataset generation, data manipulation, statistical calculations, preprocessing, and machine learning operations.

Shiny: Provides the web application framework, reactive server logic, navigation tabs, interactive controls, and user interface.

ggplot2: Used to create statistical visualizations, including histograms, box plots, scatter plots, and distribution comparisons.

Plotly: Adds interactive features to supported visualizations, such as hover information and interactive chart exploration.

DT: Provides interactive tables for viewing, searching, sorting, and examining dataset records.

Base R and stats: Support statistical calculations, descriptive summaries, correlation analysis, hypothesis testing, and logistic regression.

rpart: Supports decision tree classification.

randomForest: Supports random forest classification.

scales: Provides value-formatting utilities for the dashboard.

6. OUTPUT SCREENSHOTS WITH GITHUB LINK

6.1 Login Page

The login page serves as the entry point to the application. It includes username and password fields, a sign-in button, and validation messages for incorrect credentials. The interface uses a dedicated login layout

to distinguish authentication from the main analytical workspace.

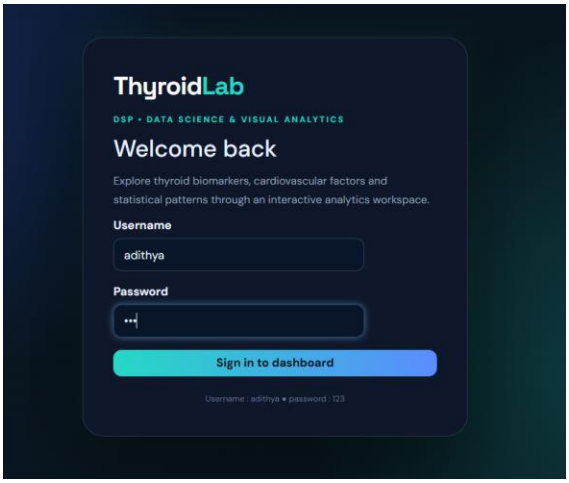


Figure 1: Insert the login page screenshot.

6.2 Overview Dashboard

The Overview Dashboard provides an initial summary of the active dataset. It displays the number of records, number of variables, count of positive synthetic cardiovascular outcome labels, and average age and BMI.

It also presents numeric-data completeness, thyroid-status composition, age distribution, BMI distribution by cardiovascular outcome, and a preview of the dataset. These visual summaries help users understand the dataset before proceeding to more detailed analysis.

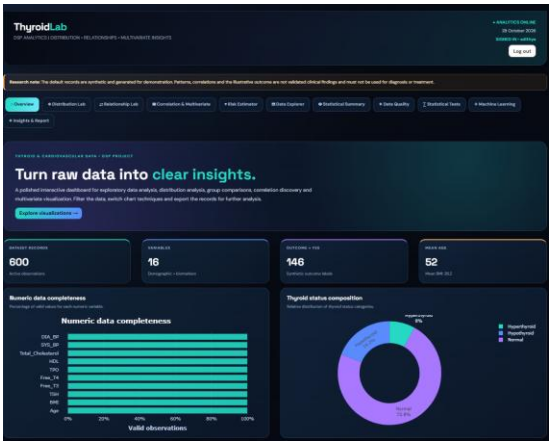


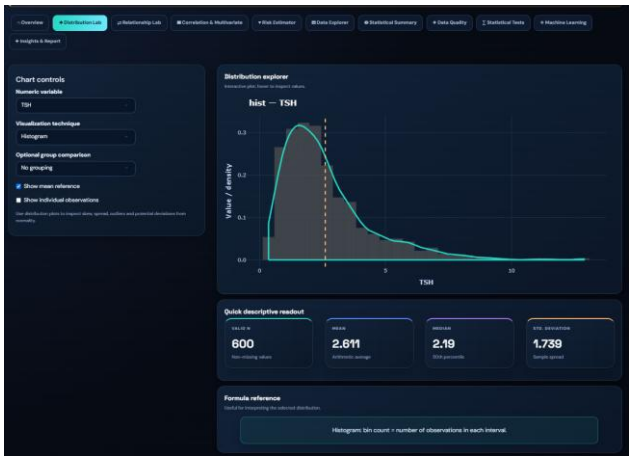
Figure 2: Insert the Overview Dashboard screenshot.

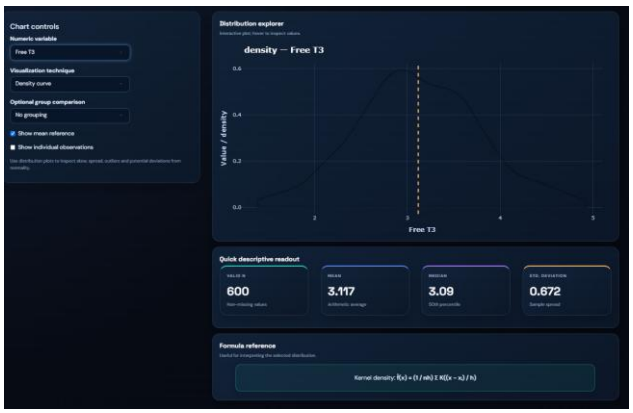
6.3 Distribution Lab

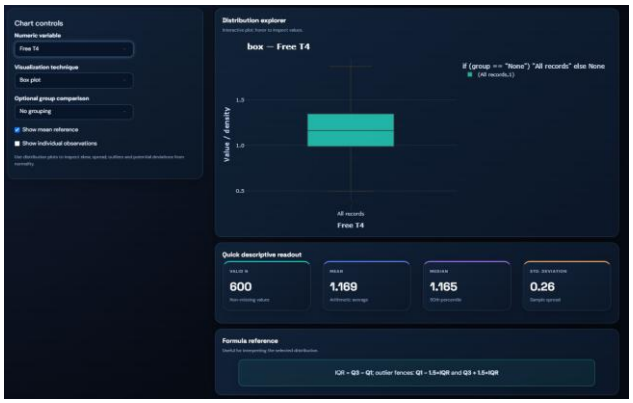
The Distribution Lab allows users to select a numerical variable and choose a suitable visualization technique. Supported options include histograms, density curves, box plots, violin plots, empirical cumulative distribution functions, Q–Q plots, rug plots, and frequency polygons.

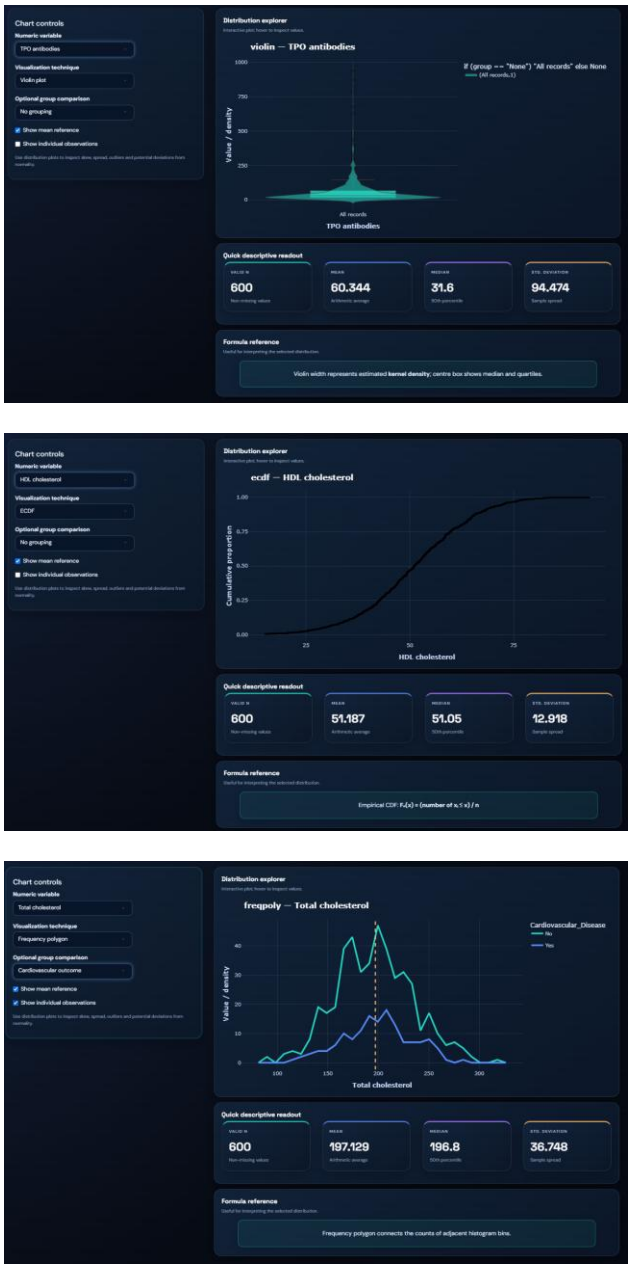
Users can optionally group observations by cardiovascular outcome, thyroid status, or gender. Additional controls allow the mean reference line and individual observations to be displayed where applicable.

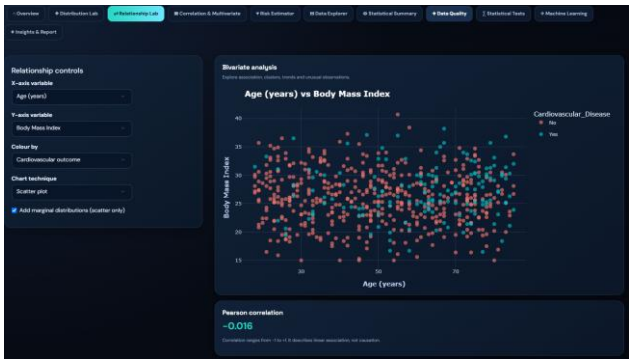
The module also displays descriptive information such as the number of valid observations, mean, median, and standard deviation.



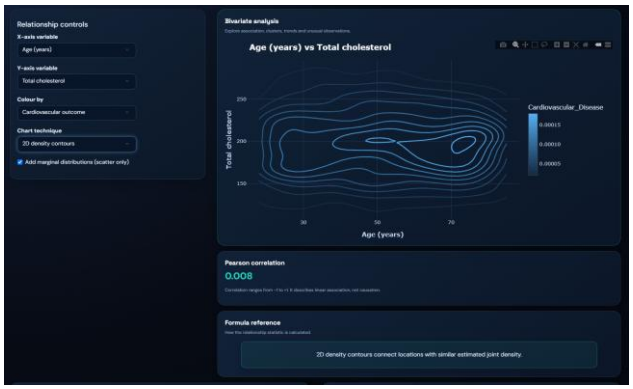












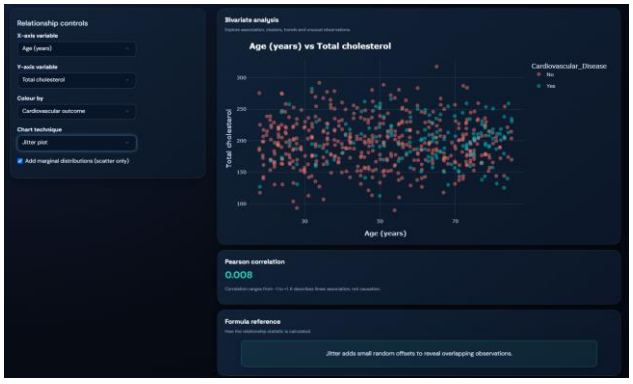


Figure 3: Insert the Distribution Lab diagrams

6.4 Relationship Lab

The Relationship Lab supports the exploration of associations between two numerical variables. Users can choose the X-axis variable, Y-axis variable, grouping category, and chart technique.

Available techniques include scatter plots, scatter plots with regression lines, binned scatter plots, two-dimensional density contours, and jitter plots. The module also contains additional comparisons involving age and systolic blood pressure and TSH and Free T4.

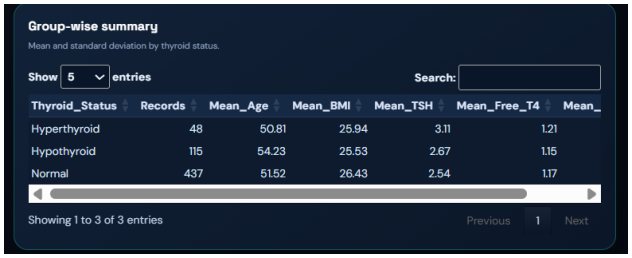


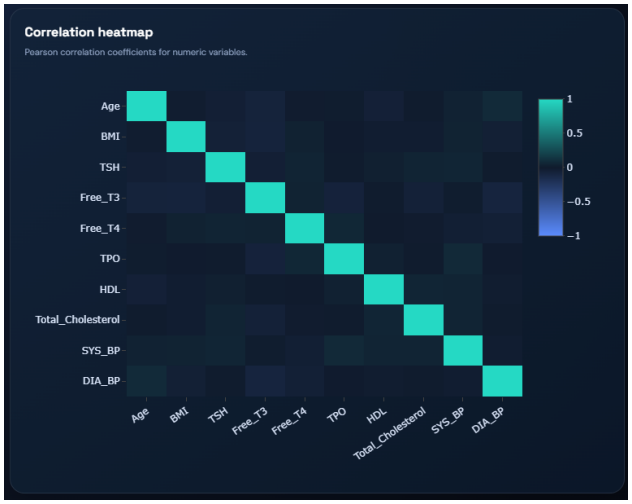
Figure 4: Insert the Relationship Lab diagrams.

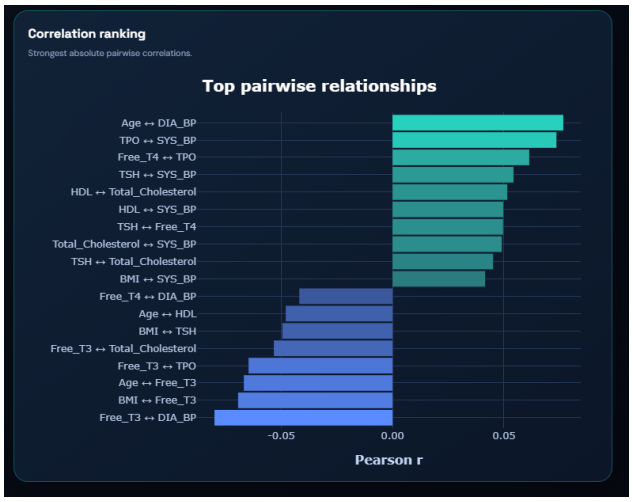
6.5 Correlation and Multivariate Analysis

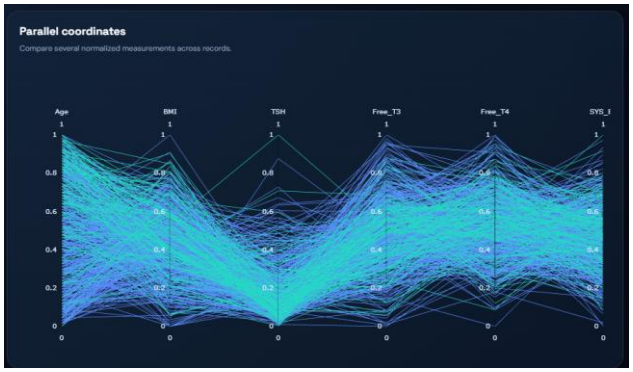
This module provides a correlation heatmap and a ranking of pairwise correlations. It also includes parallel-coordinate plots, radar profiles, pairwise scatter matrices, and group-wise summaries.

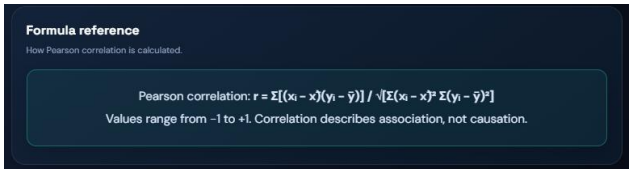
These visualizations enable users to inspect multiple numerical variables and compare the statistical profiles of different groups. The module is intended to support exploratory investigation rather than establish causal relationships.



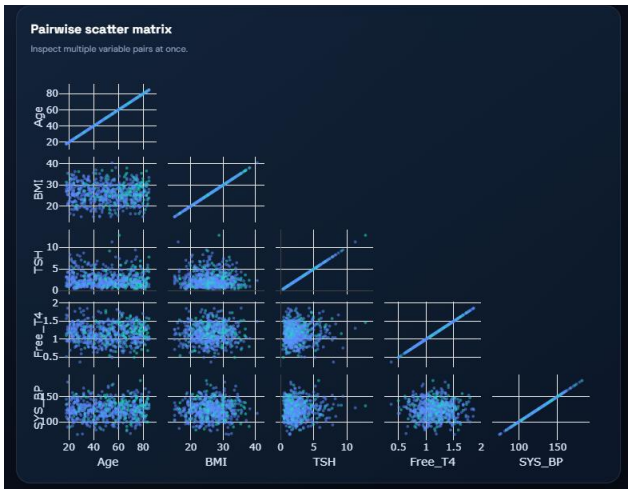








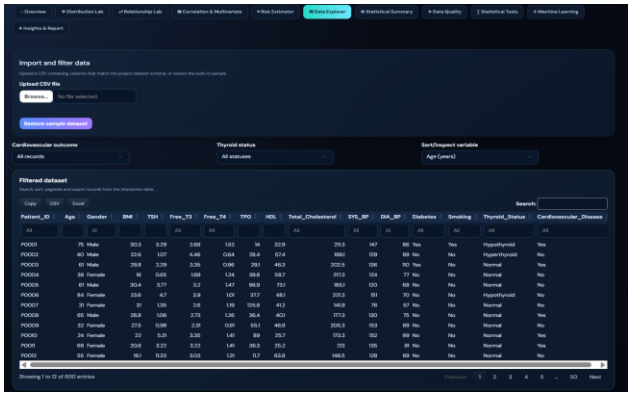




Screenshot 5: Insert the Correlation and Multivariate Analysis screenshot.

6.6 Data Explorer

The Data Explorer allows users to import compatible CSV files, restore the built-in sample dataset, filter records by cardiovascular outcome and thyroid status, inspect variables, and download filtered records.

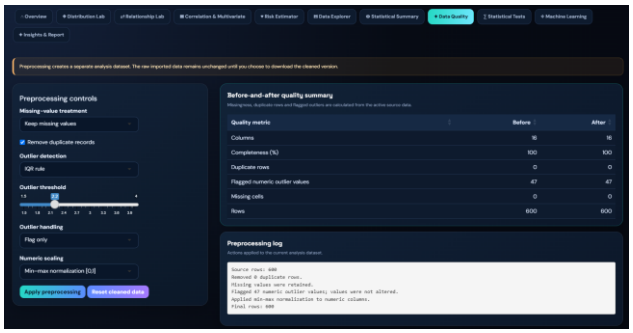


Screenshot 6: Insert the Data Explorer screenshot.

6.7 Data Quality and Preprocessing

The Data Quality module provides controls for missing-value treatment, duplicate removal, outlier detection, outlier handling, and numerical scaling.

A quality summary presents information about the source data, while the preprocessing log records operations applied to the analysis dataset. Users can inspect the cleaned-data preview, reset preprocessing, and download the cleaned CSV file.



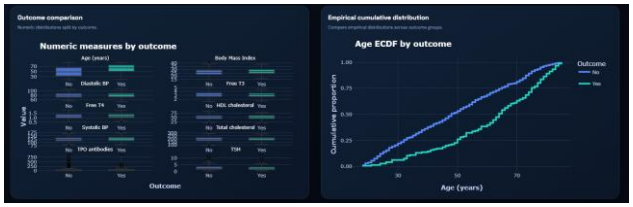


Screenshot 7: Insert the Data Quality screenshot.

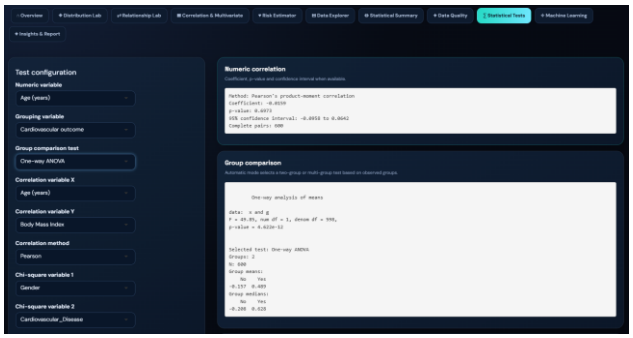
6.8 Statistical Summary and Statistical Tests

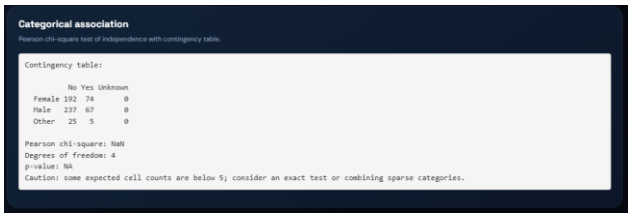
The Statistical Summary module displays descriptive statistics, missingness information, outcome comparisons, and empirical cumulative distributions.

The Statistical Tests module allows users to configure numerical correlation tests, group-comparison tests, and chi-square tests for categorical associations. Results are presented with supporting information and interpretation summaries.



Screenshot 8: Insert the Statistical Summary screenshot.







Screenshot 9: Insert the Statistical Tests screenshot.

6.9 Machine Learning

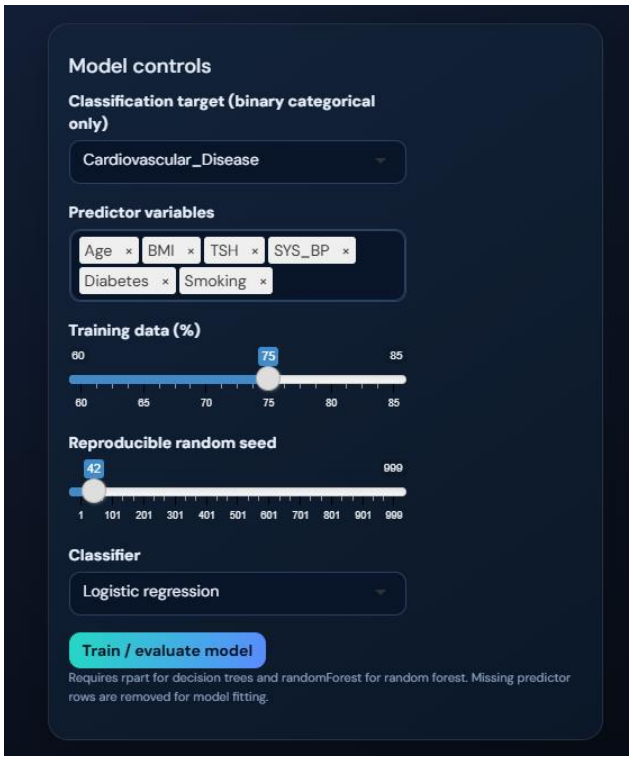
The Machine Learning module provides controls for selecting a classification target, predictor variables, training-data percentage, random seed, and classification algorithm.

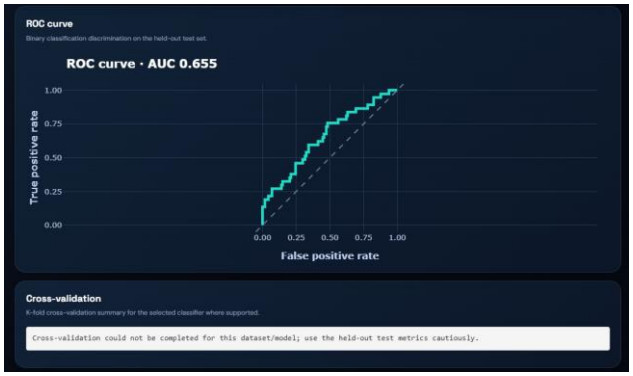
Users can select logistic regression, a decision tree, or a random forest. The module presents model-evaluation information, a confusion matrix, a feature-importance visualization where supported, an ROC curve, and cross-validation information where implemented.

The results demonstrate the model-evaluation workflow. Since the default dataset is synthetic, they should not be treated as evidence of real-world predictive performance.









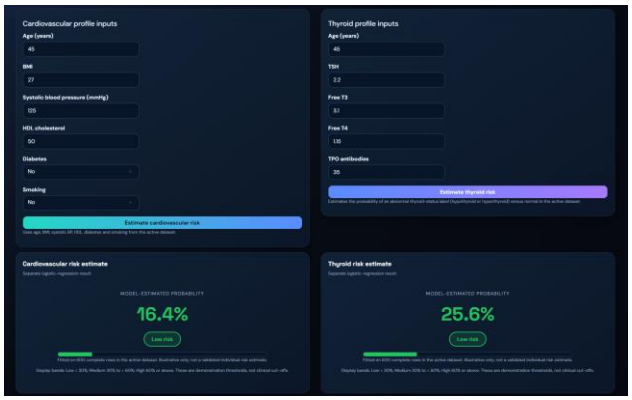
Screenshot 10: Insert the Machine Learning screenshot.

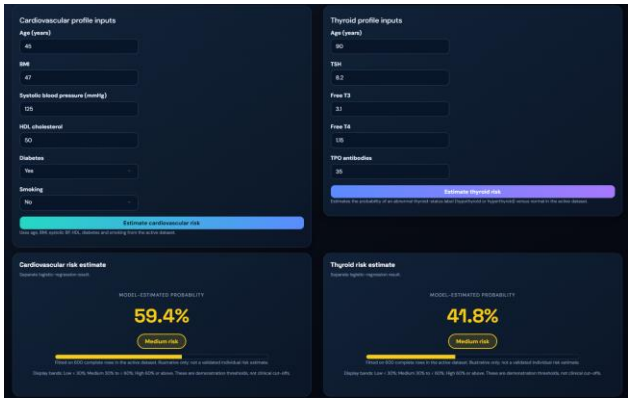
6.10 Risk Estimator

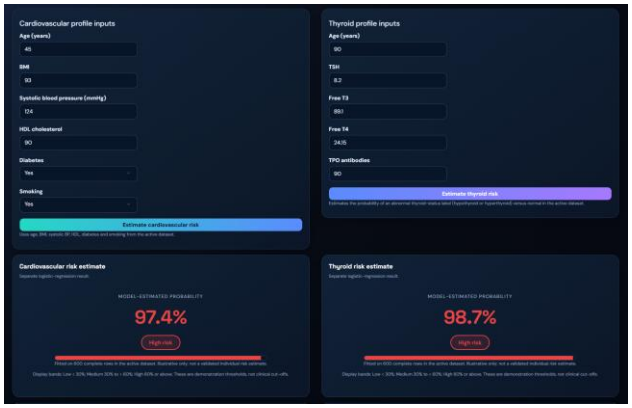
The Risk Estimator module contains separate interfaces for cardiovascular and thyroid-related probability estimates.

The cardiovascular estimator accepts age, BMI, systolic blood pressure, HDL cholesterol, diabetes, and smoking information. The thyroid estimator accepts age, TSH, Free T3, Free T4, and TPO antibody measurements.

The results illustrate how logistic-regression-based calculations can be incorporated into an interactive application. They are educational estimates and are not intended for clinical diagnosis or treatment decisions.









Screenshot 11: Insert the Risk Estimator screenshot.

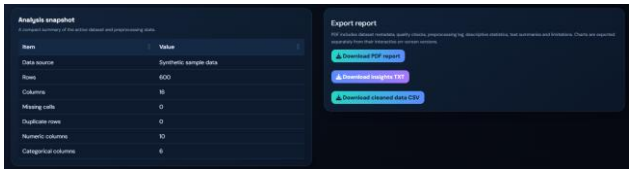
6.11 Insights and Report Generation

The Insights and Report module presents automated analytical observations and an analysis snapshot. It provides options to download a PDF report, text-based insights, and cleaned CSV data.

These features help users document their analysis and retain outputs for further examination or academic reporting.







Screenshot 12: Insert the Insights and Report screenshot.

6.12 GitHub Repository

GitHub Repository:

https://github.com/manthenasriadithya2024-design/ThyroidLab.git

7. CONCLUSION

ThyroidLab demonstrates how data science programming can be applied to develop an integrated analytical application using R and Shiny. The project combines data import, normalization, exploratory data analysis, preprocessing, statistical testing, machine learning, interactive visualization, and reporting within a unified interface.

The integration of interactive charts, data tables, preprocessing controls, model-evaluation outputs, and export facilities makes the application suitable for educational demonstrations and analytical experimentation. The project also illustrates the value of modular architecture and reactive programming when developing interactive data applications.

In conclusion, ThyroidLab provides a practical demonstration of R programming, statistical visualization, data preprocessing, hypothesis testing, and introductory machine learning. Future improvements could include validated external datasets, more extensive model validation, additional visualization techniques, and enhanced data-management capabilities. Any future healthcare-related prediction features would require appropriate clinical validation before real-world use.